# Optional Knowledge-Graph Layer

## Optional Knowledge-Graph Layer I

An optional, **LLM-gated** stage lifts the corpus from bibliometrics to hypothesis-level evidence. For each record, a local language model (Ollama, default model `gemma3:4b`) extracts structured *assertions* — each encoding a direction (supports / contradicts / neutral), a confidence score, and a short natural-language justification — against the 6 hypotheses declared in configuration. Assertions are serialized as RDF-compatible nanopublications (Kuhn et al. 2016) and scored by a citation-weighted evidence function.

# Assertion Model I

Each assertion $a$ encodes:

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- **Direction**: supports, contradicts, or neutral with respect to a hypothesis $H$
- **Confidence**: a score $c_a \in [0, 1]$ from the LLM
- **Citation weight**: $\log(1 + n_{\text{cites}})$, where $n_{\text{cites}}$ is the citation count of the asserting paper

## Assertion Model I

The evidence score for hypothesis $H$ is:

$$\text{score}(H) = \frac{\sum_{a \in A(H)^+} c_a \cdot \log(1 + n_{\text{cites}}(a)) - \sum_{a \in A(H)^-} c_a \cdot \log(1 + n_{\text{cites}}(a))}{\sum_{a \in A(H)} c_a \cdot \log(1 + n_{\text{cites}}(a))}$$

where $A(H)^+$ is the set of supporting assertions and $A(H)^-$ the contradicting ones. The score ranges from $-1$ (all evidence contradicts) to $+1$ (all evidence supports).

## Incremental Extraction I

Assertion extraction is **incremental and resumable**: assertions are appended to `nanopublications.jsonl` at configurable checkpoint intervals (default: 50 papers). On restart, already-processed papers are skipped automatically, so a long extraction run that is interrupted can resume without re-processing. The `--clear-assertions` flag discards previous results for a fresh start.

## Gating and Defaults I

This stage is optional and gated by language-model availability. With no language model configured, the hypothesis evidence scores read *pending*; with Ollama configured (as in this instance, with 127 assertions extracted), scores are populated from citation-weighted assertion extraction. The hypotheses themselves — their names and scope — come from configuration and are reported regardless of whether the scoring stage has run.

The hypotheses explored in this instance are: H1 Wakefulness Efficacy; H2 Cognitive Enhancement; H3 Low Abuse Liability; H4 Dopaminergic Mechanism; H5 Off-label Psychiatric Utility; H6 Tolerability.